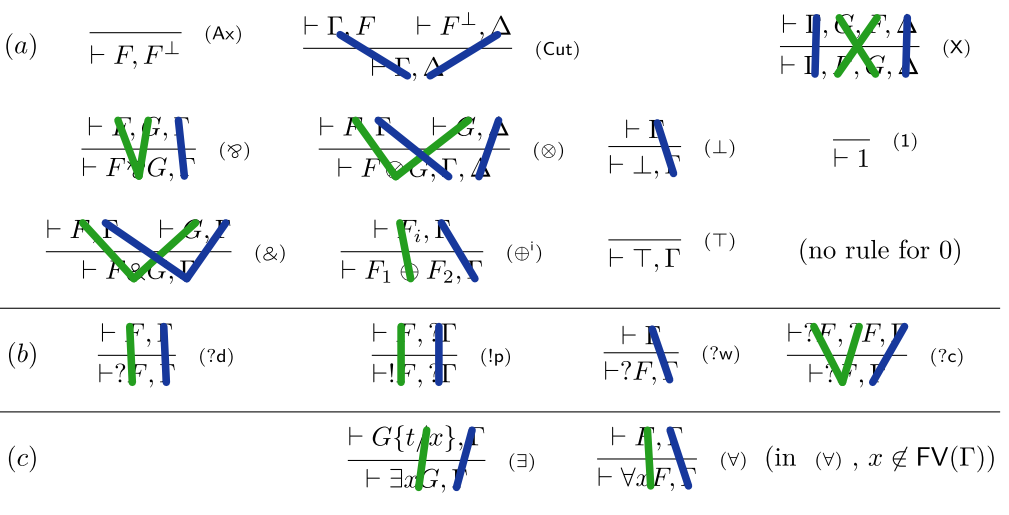
(a)
⊢ F, F⊥ (Ax)
⊢ Γ, F ⊢ F⊥, Δ / ⊢ Γ, Δ (Cut)
⊢ Γ, G, F, Δ / ⊢ Γ, F, G, Δ (X)
⊢ Γ, G, Γ / ⊢ F ⅋ G, Γ (⅋)
⊢ Γ, F ⊢ G, Δ / ⊢ F ⊗ G, Γ, Δ (⊗)
⊢ Γ / ⊢ ⊥, Γ (⊥)
⊢ 1 (1)
⊢ Γ, F ⊢ G, Γ / ⊢ F & G, Γ (&)
⊢ F_i, Γ / ⊢ F_1 ⊕ F_2, Γ (⊕^i)
⊢ ⊤, Γ (⊤)
(no rule for 0)
(b)
⊢ ?F, ?F, Γ / ⊢ ?F, Γ (?d)
⊢ F, ?Γ / ⊢ !F, ?Γ (!p)
⊢ Γ / ⊢ ?F, Γ (?w)
⊢ ?F, ?F, Γ / ⊢ ?F, Γ (?c)
(c)
⊢ G{t/x}, Γ / ⊢ ∃x G, Γ (∃)
⊢ Γ, F / ⊢ ∀x F, Γ (∀) (in (∀), x ∉ FV(Γ))